Challenge - Space Missions and Systems

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1 Part 1

Having established a state vector $X = \{x, y, u, v, GM_{gan}\}$ we consider the non linear problem of a spacecraft being attracted by three different bodies (restriced n-body problem). We assume to know exactly the position of Ganymede and Europe at every time. The dynamical problem is therefore:

$$\begin{cases} \dot{x} = u \\ \dot{y} = v \\ \dot{u} = -\dfrac{GM_{jup}}{r^3}x - \dfrac{GM_{eur}}{E^3}(x - x_E) - \dfrac{GM_{gan}}{G^3}(x - x_G) \\ \dot{v} = -\dfrac{GM_{jup}}{r^3}y - \dfrac{GM_{eur}}{E^3}(y - y_E) - \dfrac{GM_{gan}}{G^3}(y - y_G) \\ G\dot{M}_{gan} = 0 \end{cases}$$

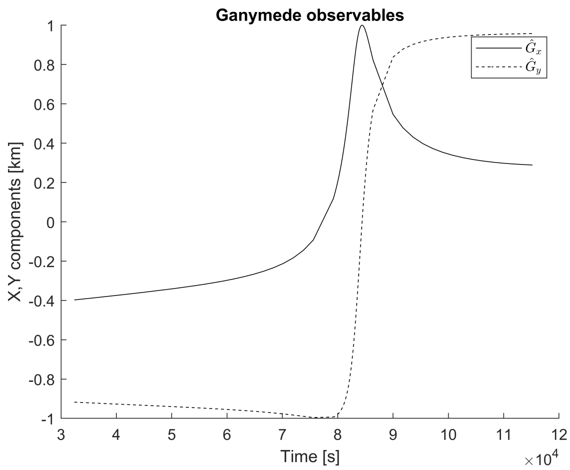
where r, E, G are respectively

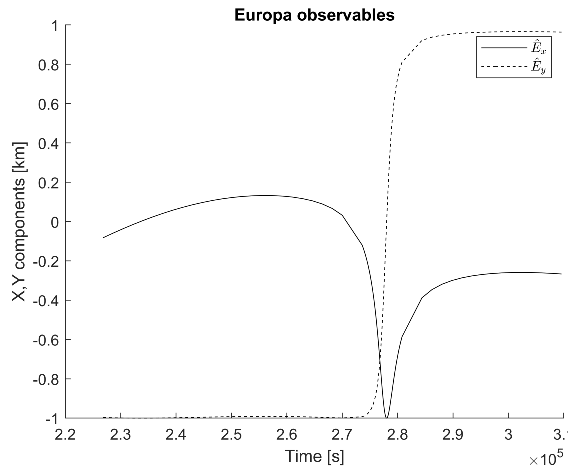
$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ E &= \sqrt{(x_E - x)^2 + (y_E - y)^2} \\ G &= \sqrt{(x_G - x)^2 + (y_G - y)^2} \end{aligned}$$

From this model we can thus calculate the linearized matrix $A = \dfrac{\partial F}{\partial X}$. We're also told to have some optical observations over different time intervals, the first one regarding the Ganymede flyby and the latter regarding the Europa one. We can obtain the observable from these data sets. Since only the components of the unit vector from the S/C to the bodies is given we'll consider them as different observables. We'll have a total of 4 observables:

$$\begin{aligned} G_1(\vec{x}, t) &= \frac{x_G - x}{\sqrt{(x_G - x)^2 + (y_G - y)^2}} \\ G_2(\vec{x}, t) &= \frac{y_G - y}{\sqrt{(x_G - x)^2 + (y_G - y)^2}} \\ E_1(\vec{x}, t) &= \frac{x_E - x}{\sqrt{(x_E - x)^2 + (y_E - y)^2}} \\ E_2(\vec{x}, t) &= \frac{y_E - y}{\sqrt{(x_E - x)^2 + (y_E - y)^2}} \end{aligned}$$

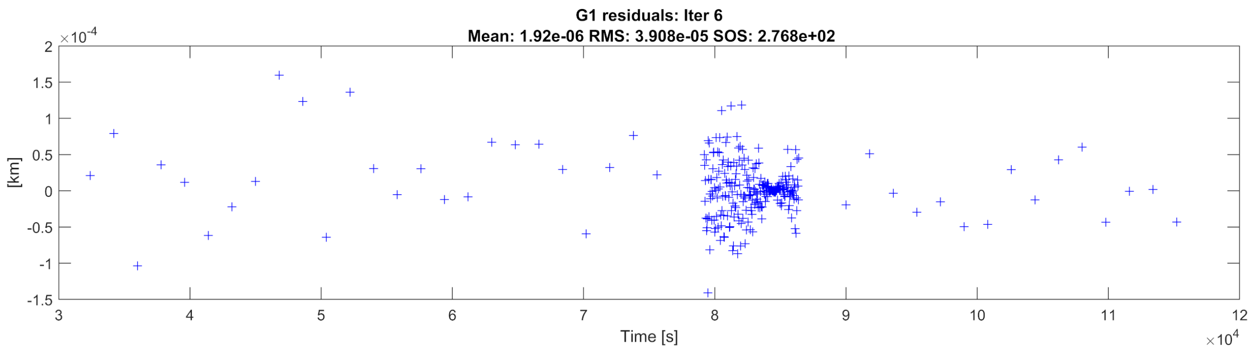
From these we can also calculate the matrix $\tilde{H} = \dfrac{\partial G}{\partial X}$.

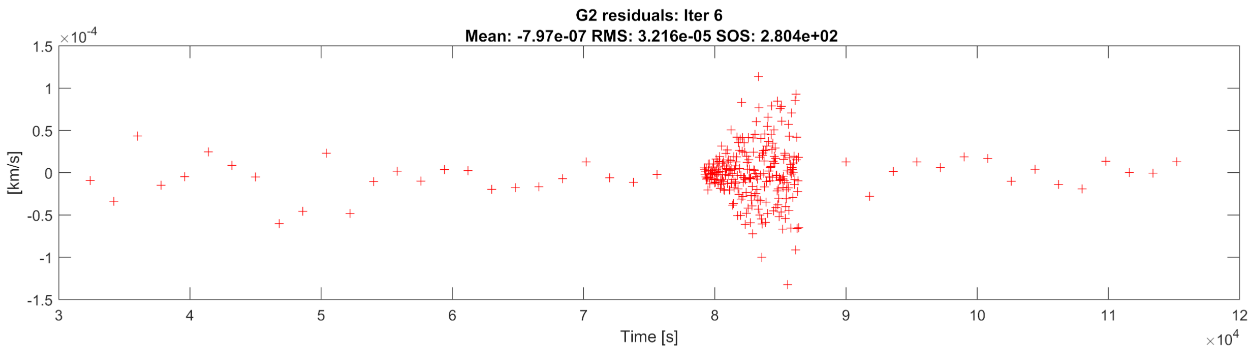




We're given a priori information about the state of the S/C alongside its uncertainties, we can use this information to run the Minimum Variance Estimator (MVE) to get a refined state at time t=0.

At first we only apply the filter with the observables that regard Ganymede (so G_1 and G_2). After running some 6 iterations we finally arrive at convergence.





We can make a SOS check and we realize this is almost exactly the number of observations (281). We can also see that the mean is almost zero. The results are the following:

$$X_0 = \begin{bmatrix} 1.1805e+06 \\ 1.2653 \\ 4.0313e-06 \\ 8.3304 \\ 9.8877e-03 \end{bmatrix}$$

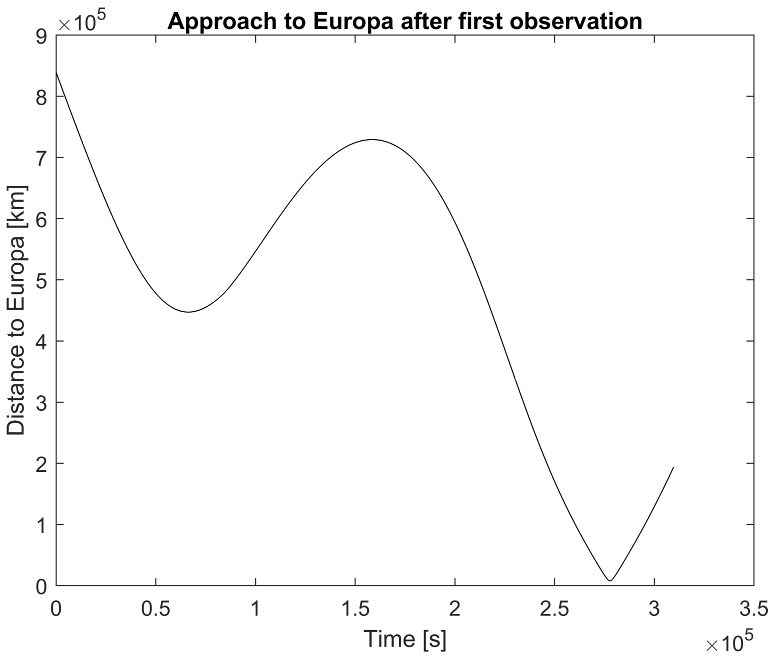
$$Pcov = \begin{bmatrix} 0.8134 & 0.0800 & -1.3122e-05 & -8.3469e-07 & 0.1985 \\ 0.0800 & 0.9579 & -2.3615e-06 & -9.7562e-06 & -0.0229 \\ -1.3122e-05 & -2.3615e-06 & 2.1314e-10 & 2.3848e-11 & -3.2752e-06 \\ -8.3469e-07 & -9.7562e-06 & 2.3848e-11 & 1.0312e-10 & 1.1366e-06 \\ 0.1985 & -0.0229 & -3.2752e-06 & 1.1366e-06 & 0.2753 \end{bmatrix}$$

2 Part 2

Using the covariance matrix calculated before we're able to integrate over time to get the estimated covariance matrix when we're at the closest approach to Europa. We need to consider that further on we integrate the less accurate the results will be, so we'll integrate over a span of 5 days.
We can then simply estimate the closest approach distance as follows and then look for the minimum in the time frame.

$$CA = |\vec{E}| = |\vec{r_E} - \vec{r}| = 7827.55\ km$$

$$t_{CA} = 2.7770e+05\ s$$



We can finally obtain the uncertainty over the distance through uncertainty propagation

$$\sigma_{CA} = \sqrt{\nabla_X{}^T P_{cov} \nabla_X} = 11.3833\ km$$

where that covariance matrix is made of the 2x2 submatrix concerning x and y, since CA depends solely on those variable. $X = \{x,\ y\}^T$ in this section.
Being CA less than 8000 km and more 7000 km, we can deduce no maneuver is needed.

3 Part 3

Using the same approach applied in Part 2 we're able to estimate the closest distance for Ganymede before and after the first observation. This time we'll integrate over a span of just 1 day, as suggested by the text.

$$CA_{\ before} = 7895.0374\ km$$

$$CA_{\ after} = 7906.1956\ km$$

with the following uncertainties

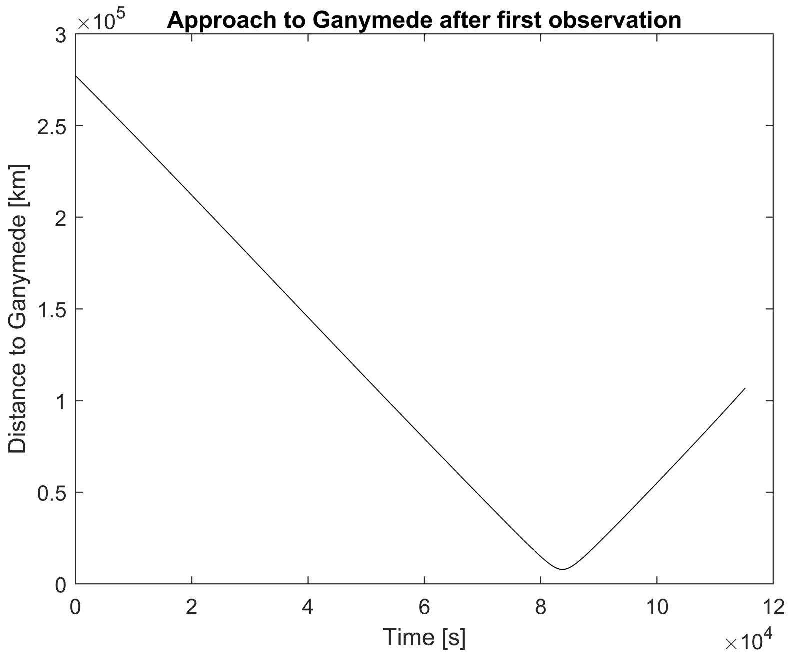
$$\sigma_{CA\ _{before}} = 91.8516\ km$$

$$\sigma_{CA\ _{after}} = 0.0601\ km$$

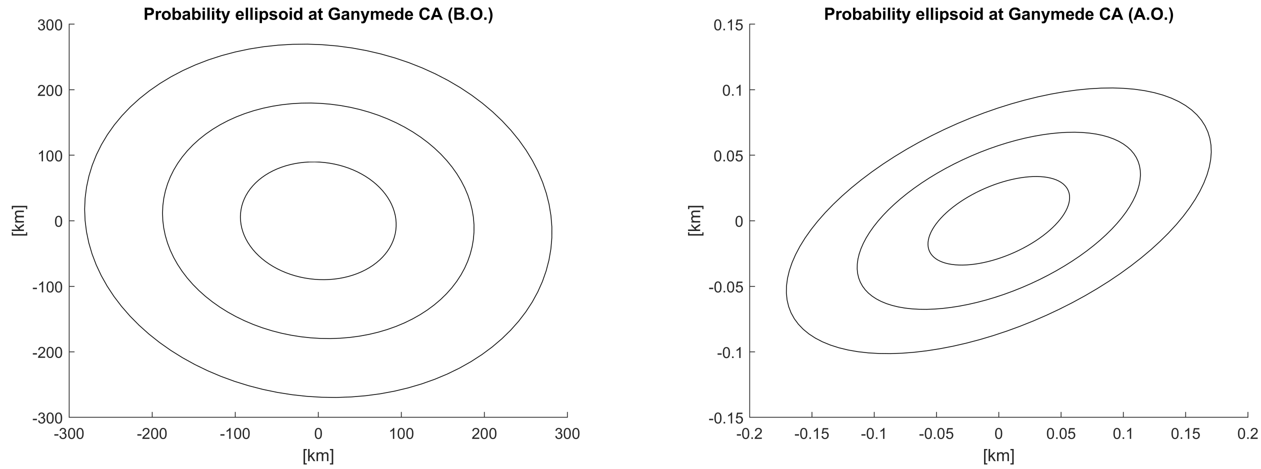
at time

$$t_{CA\ _{before}} = 8.3744e + 04\ s$$

$$t_{CA\ _{after}} = 8.3744e + 04\ s$$

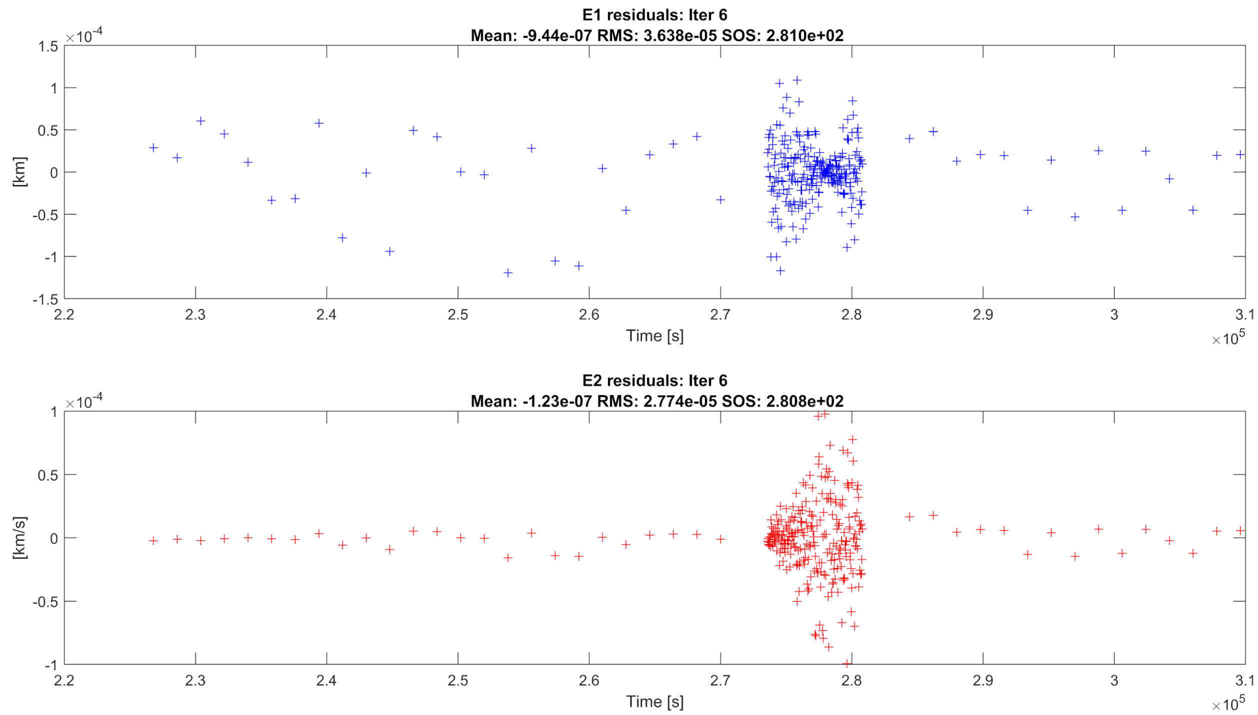


We can then draw the probability ellipsoids for the bivariate distribution (since as we've said CA depends solely on x and y) using the σ. Here only the three ellipses that encompass 1σ, 2σ and 3σ are represented. From the position of the ellipse we realize that there exists a positive correlation between x and y, as it is expected.



4 Part 4

We can then apply a second run using the observations during the Europa flyby (that means E_1 and E_2) to correct even further the state of the spacecraft at epoch t=0. Using the state at the end of Part 1 as a priori information we're able to run the MVE and arrive at convergence once more.



$$X_0 = \begin{bmatrix} 11805e+06 \\ 0.27880 \\ 1.0088e-05 \\ 8.3304 \\ 9.8877e-03 \end{bmatrix}$$

$$Pcov = \begin{bmatrix} 0.7134 & -0.2043 & -1.1179e-05 & 1.8944e-06 & 0.1436 \\ -0.2043 & 0.06178 & 3.2089e-06 & -5.6983e-07 & -0.0425 \\ -1.1179e-05 & 3.2089e-06 & 1.7532e-10 & -2.9910e-11 & -2.2738e-06 \\ 1.8944e-06 & -5.6983e-07 & -2.9910e-11 & 5.4444e-12 & 4.1544e-07 \\ 0.1436 & -0.04254 & -2.2738e-06 & 4.1544e-07 & 0.03225 \end{bmatrix}$$

5 Part 5

We estimate once again the closest approach distance to Europa. Since we are now using both data sets of observations and have successfully managed to fit the data, we can safely assume that both the σ is going to be smaller and therefore the distance more accurate.

$$CA = 7822.50\ km$$

$$\sigma_{CA} = 3.7532e-02\ km$$

6 Part 6

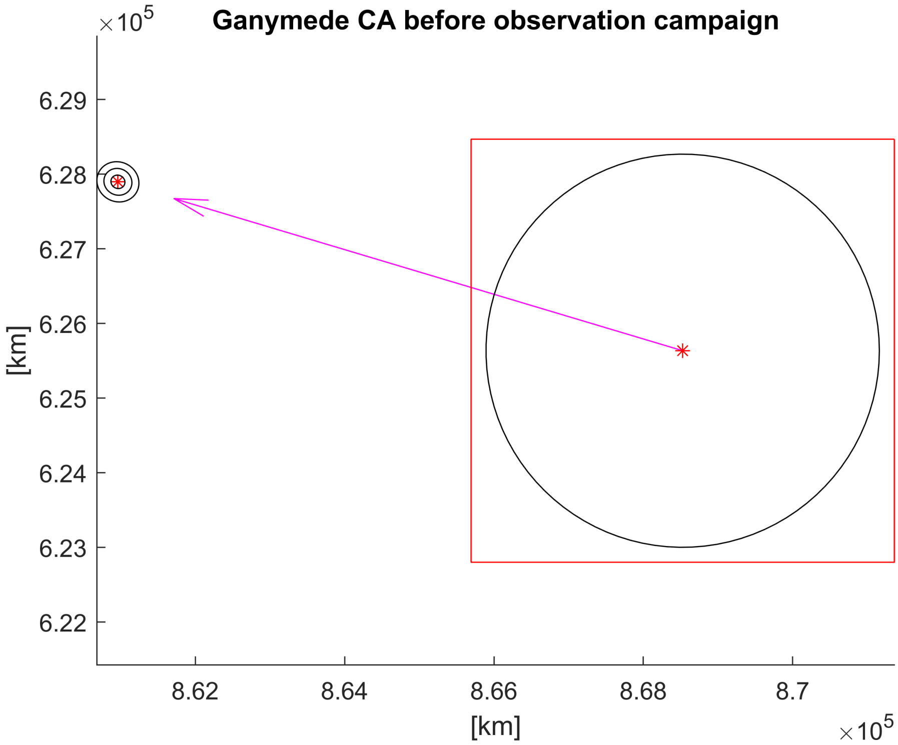
We can finally compare the two situations before and after the observation campaign, assessing therefore the probability to impact not the celestial body, but the square region of side ($R_{moon} + 400\ km$), as indicated by planetary protection rules. We have studied here the worst condition for the spacecraft during the flyby: the closest approach. For each flyby we have represented the situation as follows:

- **Moon** is the black circle
- **Square region** is represented in red around the moon
- **Probability ellipsoids** around the position of the S/C at CA

Also, the magenta arrows represents the relative position of the S/C relative to the moon, also known as the opposite of $\vec{G}$ (or $\vec{E}$). Two red asterisks are placed at the positions of the spacecraft and the center of the moon with respect to the Jupiter-centered inertial reference frame.
We are able to estimate the impact probability over that square region using the formula

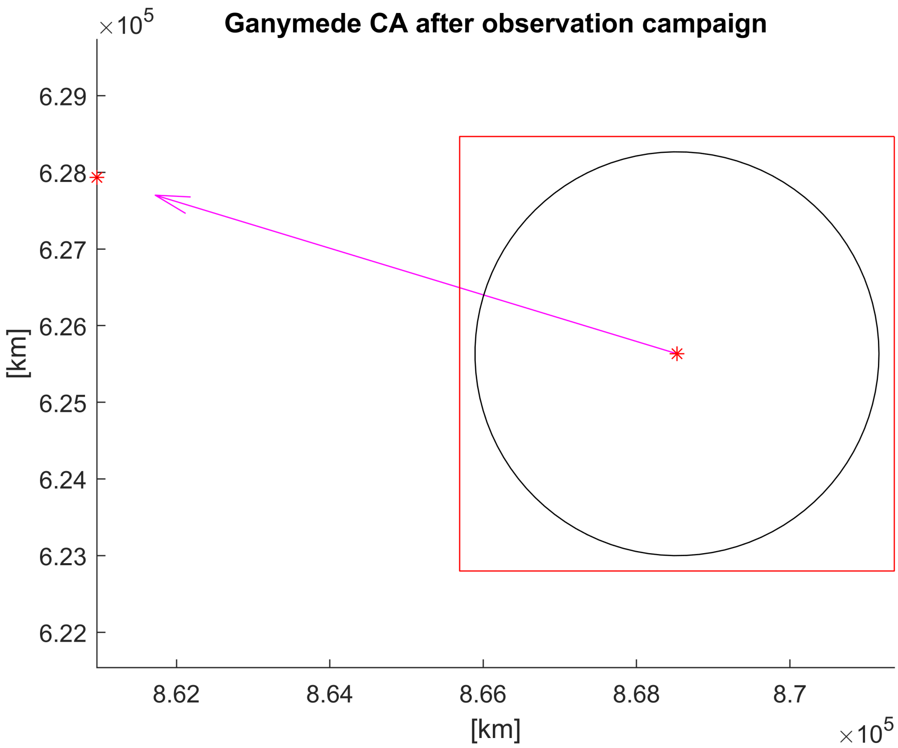
$$P = \int_A \frac{1}{2\pi\sqrt{|P_{cov}|}} \exp\left(-\frac{1}{2}(x-\mu)^T P_{cov}(x-\mu)\right)$$

where A is the area of the considered region, x is the vector consisting of the two variables (x, y), μ is the vector of the averages and P_{cov} is the 2x2 covariance matrix we considered before.
Looking at the figures we can easily recognize that the ellipsoids gets smaller after the observations are used to improve the state at time t=0. Furthermore, since we have used propagation to estimate both the covariance matrix and the distance at closest approach, it is reasonable to assume errors in the numerical method we have used. The propagation at a later time also cause the ellipsoids at CA to Europa to be larger. This eventual problem is solved hastily by using the observations.
In both cases of Ganymede's flyby we have a probability of zero to impact within the square region. The same thing cannot be said for the Europa flyby as a larger probability ellipsoid before the observation campaign yields a probability of 0.0055 to cross the protection zone. This probability is higher than the one admitted (0.005), so we might have incurred in problems, and could have called the flyby off if not for the observations, that reduces this chance to zero.



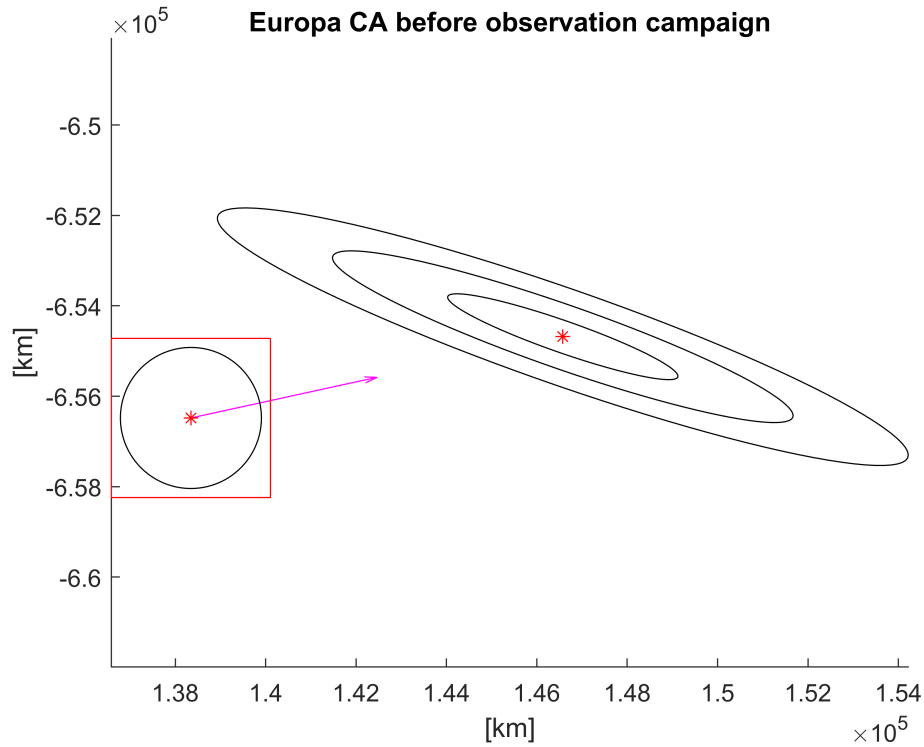
Ganymede CA before observation campaign
×10^5
6.29
6.28
6.27
6.26
6.25
6.24
6.23
6.22
[km]
8.62
8.64
8.66
8.68
8.7
[km]
×10^5

$P = 0$

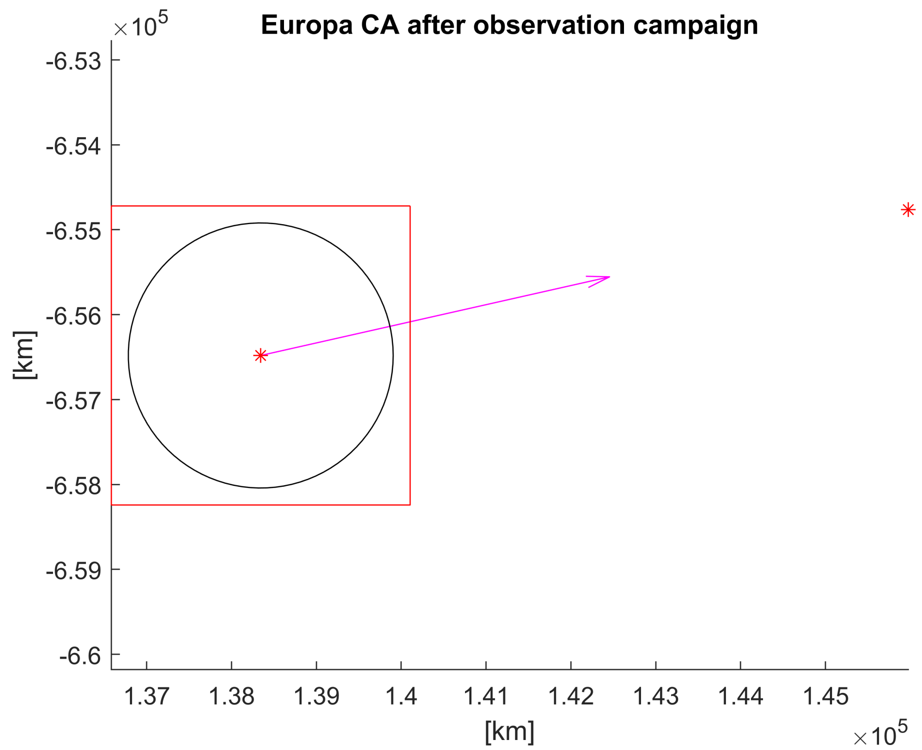


Ganymede CA after observation campaign
×10^5
6.29
6.28
6.27
6.26
6.25
6.24
6.23
6.22
[km]
8.62
8.64
8.66
8.68
8.7
[km]
×10^5

$P = 0$


Europa CA before observation campaign
×10^5
-6.5
-6.52
-6.54
-6.56
-6.58
-6.6
[km]
1.38
1.4
1.42
1.44
1.46
1.48
1.5
1.52
1.54
[km]
×10^5

$$P = 0.0055$$


Europa CA after observation campaign
×10^5
-6.53
-6.54
-6.55
-6.56
-6.57
-6.58
-6.59
-6.6
[km]
1.37
1.38
1.39
1.4
1.41
1.42
1.43
1.44
1.45
[km]
×10^5

$$P = 0$$